The Quiet Handover

What organisations give up when they deploy AI — and how to keep it

On agency: who decides, who judges, and who is still accountable when the work writes itself.

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1. The decision nobody made

Ask most organisations when they decided to let a machine make product decisions, and they will tell you they never did.

They are being honest. Nobody convened a meeting, weighed the options and resolved that from Q3 onwards a language model would settle questions of scope, behaviour and architecture on the organisation's behalf. There was no paper, no steering group, no risk assessment. What there was instead was a licence rollout, a quiet improvement in tooling, a few teams getting noticeably faster, and a slow drift in where the thinking actually happened.

That is the shape of it. Agency is rarely surrendered. It leaks.

Like many, I spent the best part of eighteen months approaching AI tool-first. Licences were granted, tooling was provided, and the measuring began. When the rate of delivery stubbornly refused to match the speed of code generation, my peers and I were left scratching our chins. The productivity potential was obvious — you could watch it happen on screen. Actually getting change in front of customers still felt slow.

It took me longer than it should have to see why. **AI does not empower evenly, and it does not empower at all without direction and intent.** Believe the headlines, the ones describing a black box with a *go faster* button on the side, and you stop paying attention to the one variable that actually determines the outcome: the people behind the wheel. You have bought an extraordinarily fast car for an organisation where nobody has been taught to drive.

But there is a second failure folded into the first, and it is the one I now think matters more.

Tool-first rollouts tend to arrive wrapped in rules — approved models, permitted use cases, prohibited data, a policy published before anybody has built up enough experience to know what the policy should say. The instinct is understandable and the intent is good. The effect is to foreclose the exploration through which people would have learned how the thing actually behaves: where it is strong, where it quietly fails, what it is worth trusting with. We hand people a technology nobody fully understands yet, and then remove their licence to find out.

That is the first agency we take away, and it is the one that would have earned all the rest back.

It also explains the most useful finding in the enterprise literature. MIT's NANDA report set out to understand why so few enterprise AI projects were producing anything measurable. Reviewing more than 300 publicly disclosed initiatives, interviewing 52 organisations and surveying 153 senior leaders, it found that roughly 95% of the custom, task-specific tools it examined had delivered no measurable return at all. That figure has since been flattened into "95% of AI fails", which is not what it says and not what it measured. The diagnosis underneath it is the durable part, and it is the sentence worth carrying out of the report: "*The core barrier to scaling is not infrastructure, regulation, or talent. It is learning.*" (Challapally et al., 2025). It is not peer-reviewed and the authors describe

their own figures as directionally accurate, so take the number loosely. The direction is what matters here, and it points at the thing a rules-first rollout prevents.

This matters more than the productivity debate everyone is having, because productivity is recoverable and agency largely is not. A team that adopts AI badly and gets slower can stop and try again. A team that has lost the ability to explain why its system behaves the way it does cannot simply reverse out. The reasoning was never written down, the people who held it have moved on, and the code was generated from a prompt nobody kept. The knowledge is not somewhere else. It is nowhere.

The risk is not that AI makes bad decisions. It is that it makes acceptable ones, continuously, on questions you did not realise were being decided.

This paper is about that handover: how it happens to individuals, how it then happens to organisations, and what it takes to keep the parts worth keeping. It is a practitioner's account rather than a philosophical one. I am interested in where the decisions actually go.

The position I will argue is not that AI should be held at arm's length. It is that the level of autonomy you grant is a design decision, it is currently being made for you by product defaults, published policy and individual habit, and making it deliberately is one of the few genuinely high-leverage things a delivery organisation can do right now.

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2. What we actually mean by agency

Agency is one of those words that sounds precise and is not. Before arguing about losing it, it is worth saying what there is to lose.

I find it useful to break it into four distinct things, because they erode at different speeds and in that order.

	What it means	How you know it has gone
Authorship	You made the thing, or directed its making closely enough to call it yours	You cannot account for why it is the way it is
Judgement	You can tell whether the output is right, and you actually exercise that	Review becomes approval; "looks fine" becomes the verdict
Control	You can change the system's direction, and change your mind later	Options have quietly closed; the tooling, vendor or structure now decides
Accountability	You own the consequences	Unchanged — and that is the problem

The uncomfortable asymmetry is in the last row. Authorship, judgement and control can all slide away. Accountability does not move at all. Nobody has ever explained an outage by saying the AI wrote it, and nobody ever will.

So the risk is not a clean trade of responsibility for speed. It is a widening gap between what you are answerable for and what you actually directed. Everything that follows is an account of that gap opening up.

One thing sits underneath all four, and it is where this paper started: **the freedom to explore**. Judgement is not issued with a licence; it is built by using something badly, noticing, and adjusting. An organisation that constrains exploration before anyone has learned anything has not protected itself — it has prevented the judgement it will shortly need. That strand returns later in this paper, where the evidence on what people do under restriction is uncomfortably clear.

A second distinction worth making early, because the whole argument turns on it. There is a difference between *delegating a task* and *delegating a decision*. Asking a model to implement a well-specified behaviour is delegation of work. Asking it to build a feature from a two-line ticket is delegation of judgement, because every question the ticket failed to answer gets answered anyway — just not by you. The two look identical from the outside. Both produce a merged pull request and a green pipeline.

PART I — HOW INDIVIDUAL AGENCY GOES

3. The handover is gradual, which is why nobody signs it off

Nobody moves from writing code to supervising an agent in a single step. The handover happens in increments small enough that none of them feels like a decision.

There is a measured version of this, and it is the most useful single fact I know on the subject.

Anthropic looked at their own usage data for one specific task: producing a blog post or article. Same kind of request, same models available. The only variable was which product the person used.

The two differ in a way worth spelling out. In the chat products, you type, it replies, and nothing further happens until you read the reply and respond — you are in the loop by construction. Claude Code is an agentic tool: it runs in the terminal with access to your files, so a single instruction can set off a long chain of reading, writing and running things before it reports back. The loop closes without you.

In chat, the median session producing an article took **thirteen rounds of back-and-forth** — the person wrote something, read what came back, pushed, corrected, redirected, and arrived at the result jointly. In Claude Code, the median session producing the same kind of artefact contained **a single human prompt**. One instruction in, finished work out.

Same task. Same models. Thirteen interventions, or one, depending entirely on where it was typed.

The obvious objection is that Claude Code users are running stronger models — and they are, with 54% of sessions on Opus against 10% in chat. But the gap survives the controls. Comparing only sessions served by the same model, Claude Code still shows measurably more autonomy. Anthropic's own conclusion: "*the product used is likely more important than the underlying model.*" (Anthropic, 2026b)

Read that with agency in mind. Nobody in the Claude Code group decided to care less about the outcome. They simply worked in a place that did not ask them thirteen times what they thought.

The level of delegation is set by design decisions, not by user intent.

The wider trend points the same way, with a genuine wrinkle that has to be reported honestly. Directive delegation, meaning handing over a complete task rather than collaborating through it, rose from 27% to 39% between December 2024 and August 2025, tipping usage into automation territory for the first time. It then partially reversed by November 2025, which Anthropic attribute to product changes (Anthropic, 2025b; 2026a). So the honest reading is volatility with a rising ceiling, rather than a one-way march.

The reversal actually strengthens the argument rather than weakening it. If the amount of control a person exercises moves up and down with product release notes, then it is not the person setting it.

And the direction of travel is clear enough in practice. Each increment is rational on its own terms. You stop reading the diff line by line because the last forty were fine. You stop writing the ticket in detail because the model infers it well enough. You stop asking why a library was chosen because the choice was reasonable. None of these is wrong. Collectively they are a transfer of authorship, and there is no moment at which anyone decides to make it.

Autopilot

Aviation worked all of this out decades ago, at considerable cost, and the vocabulary it developed is the one we need.

Autopilot was not introduced to remove the pilot. It was introduced to remove the tedium — the constant small corrections that consumed attention without requiring skill. It worked. Flight became safer, longer and far less exhausting. And it produced a problem nobody designed for: a pilot who is present, qualified, and no longer flying the aircraft. The hands come off, the scan gets shallower, the questions stop being asked, and the loss of practice is invisible precisely because nothing is going wrong.

That is the shape of what AI tooling does to a delivery organisation, and the mechanism is convenience rather than complacency. Nobody decides to stop thinking. The output arrives already formed, already plausible, already in the right shape, and interrogating it now costs more than accepting it does.

What makes this hard to catch is that every instrument reads normal. Code is shipping faster. Tickets are closing. The pull requests are well-formed and the tests are green. We grow comfortable with the quality, and comfort does the rest: the pause before pressing enter gets shorter, and then disappears. We stop asking how the thing reached its conclusion, or why it chose this approach over the three it discarded without mentioning.

And the higher you sit, the better it all looks. This is the altitude problem, and it is the reason the erosion goes unchallenged for so long. From the flight deck you might notice that your hands have not touched the controls in an hour. From the ground, all you can see is an aircraft holding a perfect heading. Delivery metrics are a view from the ground. They report that the work is moving; they are silent on whether anyone is still flying it.

The trouble arrives at the handback. Something goes wrong. An incident, an audit, a regulator, a customer asking why the system did what it did. Control returns to a human at the exact moment the situation is hardest, and they have been away from the controls for months. Bainbridge named this in 1983: automation leaves the operator the residue it cannot handle, while removing the practice that would have made them capable of handling it (Bainbridge, 1983). Section 5 is about what that does to review; this is the version that does it to a whole organisation.

One instrument has been flashing for two years

There is a measure that catches this, and it has been pointing the same way since 2024.

DORA's annual research tracks what it calls **delivery stability** — in plain terms, how often the things you ship turn out to be broken. It combines change failure rate (the share of deployments that cause a problem in production and need a hotfix or a rollback) with rework rate (deployments you did not plan, made to fix a user-facing bug). It is, roughly, the measure of whether the work was actually right.

In 2024, DORA found that **for every 25% increase in AI adoption, delivery stability fell by an estimated 7.2%.** Throughput barely moved, an estimated 1.5% reduction over the same interval (DORA, 2024). The 2025 study repeated the exercise with a differently constructed measure of AI adoption and found the same direction again, noting plainly that "*AI is associated with an increase in software delivery instability*", though it reports this only as a position on a chart, without publishing a comparable percentage (DORA, 2025). Both are self-reported survey data rather than telemetry from production systems, and both are correlations. DORA says so itself, and it matters here, because a correlation this convenient to my argument deserves more scepticism than one that is not.

Read that back slowly, because it is not the finding people assume. The problem is not that less is being produced. It is that **more of what gets produced turns out to be wrong.**

I do not think this is AI writing bad code. I think it is good-enough code going out unexamined. Every mechanism in this section points that way: output that arrives faster than judgement can be applied to it, a confident register that gives reviewers no reason to pause, and an organisation measuring the volume rather than the scrutiny. A change failure is what happens downstream when nobody asked the question upstream. The models are not the variable that changed most between 2023 and now. The amount of human attention per line shipped is.

DORA's own 2025 framing is consistent with this, and I would put weight on their phrasing rather than mine: instability rises "*unless the surrounding system and culture changes.*" The tool is not the determining factor. What the organisation does around it is.

That is a reading of the correlation rather than a finding, and the same data fits other explanations: larger changes, less familiar code, or teams attempting work they would previously have declined. It is worth testing against your own incident record before you accept it from me.

Autopilot is not the problem. Autopilot with nobody watching the instruments is the problem — and the one instrument that does measure it has been telling us so for two years.

One qualification, which section 6 picks up properly: none of this is a story about weak-willed individuals. The slide is made rational by the organisation around it.

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4. From producing the work to directing, judging and owning it

Let me be careful here, because this is the point at which this kind of argument usually goes wrong.

The first move is an identity one: from "*I produce the work*" to "*I direct, judge and own the work.*" **That move is not the problem.** It is probably right, it is certainly happening, and a paper that treats it as a loss to be mourned is no use to anyone. Most of the people I work with would take it willingly.

It is harder than it sounds, though. If your sense of a job well done is tied to the green tick on a pull request you laboured over, the tick still arrives — it just arrives without the labour, and often without the satisfaction. The artefact is identical. The feeling is not.

That discomfort is legitimate, and treating it as resistance to be managed away is a mistake. People who believe a change is not in their interest rarely argue about it. They find quieter ways to make sure it does not succeed.

Direct, judge and own is three distinct jobs, and most of us were only ever trained for one.

The craft does not disappear. It moves — from typing to taste.

The problem is not the transition. It is how we run it.

Here is the thing we get wrong, and I have got it wrong myself.

We take people whose professional identity is built on X — on producing the work, on being demonstrably good at it, on a definition of value they have spent a decade earning. We tell them the job is now Y. And then, in the same breath, we tell them *how* to do Y: these models, these use cases, this prompt library, this approved pattern, this review template, this policy.

Every part of that is well-intentioned. Collectively it is the same foreclosure as section 1, arriving a layer down. We have replaced the thing someone was good at with a thing nobody has yet learned to be good at, and then removed the only route by which they could have learned it — using it, badly, repeatedly, and noticing what happened.

Telling someone their value now comes from judgement, while prescribing the judgement, is not a transition. It is a demotion with better branding.

It does not work on its own terms either. Directing moves the effort upstream: the hard part is no longer building, it is framing the problem precisely enough that what comes back is worth having. That is a specification skill, identical whether you are briefing a contractor, a junior colleague or a model — and it is learned the same way, by watching your own intent come back misread and working out which part of your framing caused it. Most people have never had to make their intent explicit, because they were the ones executing it. The context lived in their head and never needed to leave. You cannot hand that over in a training deck. It is acquired by being wrong in public, cheaply, often.

Owning does not move at all. Accountability stays exactly where it was — worth saying early and often, because it is also the answer to the question nobody asks aloud: *am I training my replacement?* If the accountability stays with you, so does the role. It just becomes a role defined by judgement rather than output.

Which leaves judging. It gets its own section, for a reason that is now the argument rather than an aside.

The precedent

There is a precedent worth knowing, because it shows a profession getting this wrong from the other direction. Nelson and Irwin studied librarians facing internet search. The first reaction was not curiosity but distance: search was the shallow option, librarians were the people who found things properly. The profession defined itself *against* the technology instead of engaging with it, and lost the best part of a decade before rebuilding the role around the new tools (Nelson and Irwin, 2014).

They were not foolish. They judged search by the standards of their own craft, where it genuinely was worse — and being good at something is exactly what makes it hard to see the thing about to change it.

Two failure modes, then, and they arrive at the same place. **A profession can refuse the transition, and a management team can prescribe it so tightly that nobody learns to make it.** Both end with people nominally responsible for judgement they were never given the room to develop.

That is the setup for what follows. We have moved people onto judgement, and skipped the part where judgement is built.

This section argues that the transition is right and that prescribing it is the failure. It leaves the other half open: how someone is actually helped to find where they now add value. That is section 13, and the answer there is not a curriculum.

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5. Judgement is the first thing to go

Of the three jobs, judgement is the one we assumed would take care of itself. It was the part people were already doing (reviewing, approving, having an opinion), so it did not look like a new skill that needed building. That assumption is wrong twice over: the task itself got materially harder at the same moment we started relying on it more, and nobody was equipped for the harder version.

Reviewing generated work is harder than reviewing your own, and we have been consistently pricing it as though it were easier.

When you review code you wrote, you are checking an artefact whose reasoning you still hold. When you review code a model wrote, the reasoning does not exist anywhere. You have to reconstruct it from the output, which is precisely the harder direction of travel. Lisanne Bainbridge named this in

1983 as one of the *ironies of automation*: automating the easy parts leaves humans the harder residue, while eroding the practice that made them good supervisors in the first place. Forty years and two thousand citations later it reads like a description of a code review (Bainbridge, 1983).

Then add automation bias; our documented tendency to accept a confident machine answer more readily than an equivalent human one (Parasuraman and Manzey, 2010).

Two things make this worse in practice than the literature suggests.

Confidence is not calibrated — the model does not sound less certain when it is less certain. Plausible and correct look identical on the screen. There is no tell, no hesitation, no "I'm not sure about this bit". The register is the same whether the model is reproducing a well-trodden pattern or inventing a package that does not exist (Spracklen et al., 2025).

Volume defeats attention. Review capacity is the one input that has not grown. Across 49,000 developers, 66% named "*almost right, but not quite*" as their single biggest frustration with AI output (Stack Overflow, 2025) — and "almost right" is exactly the category that survives a tired review at 4pm on a Thursday. The failure mode is not accepting obvious rubbish. It is accepting the near-miss.

The result is a quiet redefinition of what review means. It stops being an act of judgement and becomes an act of approval. The ceremony is intact. The pull request, the approver, the audit trail. What has evaporated is the judgement all of it was supposed to encode.

A rubber stamp and a considered approval are indistinguishable in the git history.

There is no good published evidence on what review should look like when the thing under review was generated. I have looked. There is an established literature on code review, and an established literature on automation bias, and almost nothing on the intersection — which means anybody offering you a review process for AI-generated code is offering you their experience, including me, later in this paper. Worth knowing before you adopt one.

What I will say here is only the diagnostic half: **review is now the load-bearing control, and it is the one we have changed least.** The practices I would put against that are in section 14, where reviewing the specification rather than the diff does most of the work.

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6. Skill follows judgement out of the door

Judgement is a skill, not a possession. Exercise it less and you have less of it. Use it or lose it, in other words; and this is not a conjecture about the future. It is the oldest finding in the automation literature, and there is now direct evidence in our own domain.

Start with the finding that transfers. Perry and colleagues at Stanford ran a controlled study in which 47 participants completed five security-related programming tasks across three languages, with 33 given an AI assistant and 14 working unaided. The group with the assistant wrote measurably less

secure code. I would not lean on that half of it: the assistant was a 2022-era model, and anyone who wants to dismiss the result can reasonably say the tools have moved on.

The more relevant finding is the other one. **The assisted group was more confident its code was secure, at the same time as being less likely to be right.** That is not a fact about one or another model's capability; it is a fact about what happens to a person's self-assessment when a fluent answer appears in front of them. Models improve. That effect does not go away, because it was never about the model.

There is a detail in the same study that points directly at the remedy. Participants who trusted the assistant less, and who worked harder at their prompts (rephrasing, adjusting, pushing back), produced code with fewer vulnerabilities (Perry et al., 2023). **The people who treated the output as a draft to argue with did better than the people who treated it as an answer.** It is the one piece of direct evidence I know of that retained agency shows up in the artefact. It is also one study, 47 people, only 14 of them in the control group and a good number of them students. Not enough to prove anything, enough to be worth attention, and I have not found better.

Small as it is, it describes something I recognise, and I suspect you will too. Think of two people on your own team: the same tooling, the same task, the same deadline. One of them argues with what comes back — reads it properly, questions the approach, sends it round again. The other reads it, finds nothing obviously wrong, and ships it. You can probably name both of them without much effort.

Recognition is not evidence, and a small study plus a room of nodding heads is still not proof. But it is worth something that the measured effect and the lived one point the same way. What neither of them explains is why. **Why do some people hand over their agency, often without noticing they have done it, while others keep pushing back as we would hope they would?**

There is at least a partial answer, and it is not the one most people assume. Lee et al. (2025) surveyed 319 knowledge workers and found that higher confidence in generative AI was associated with *less* critical thinking, which is unsurprising. The useful half is the mirror image: **higher confidence in one's own skill was associated with more of it.** The people who kept challenging were not the ones who distrusted the AI. They were the ones who trusted themselves. Self-report about one's own thinking is a weak instrument and the study is a snapshot rather than a tracked change, so this is corroboration rather than proof. But it points somewhere uncomfortable for anyone running an adoption programme: **the variable you can actually move is not how much people trust the tool. It is whether they still believe their own judgement is worth applying.**

Nobody chooses the path of least resistance in a vacuum

It would be easy to read all of that as a story about individual character. Some people push back, some people do not, and the difference is temperament. I do not think that is what is happening, and I think the organisation is doing more of the work than it realises.

Look at what we actually set up.

We buy the licences and hand out the tools. We have read the same posts as everyone else, the ones promising tenfold gains and a weekend of work done in an afternoon, and we bring that expectation into conversations about delivery, usually without saying so out loud.

The real numbers were there all along, and section 11 sets them out properly. Credible measured uplift clusters around 20–30%, and the range includes a measured *slowdown*. **No controlled study has found anything close to tenfold.**

But teams are not managed against the studies. They are managed against the marketing. And the expectation does not arrive as a claim someone can check and push back on. It arrives as pressure.

Then we make it worse by changing our minds.

The gains do not show up in the first quarter, so we switch tools. Then we change the framework. Then the approved model changes, or the platform team rolls out a new way of working, and everyone starts again. Each move is defensible on its own. Together they are corrosive, because **the thing being reset every time is the only asset that was actually accumulating: people getting good at this.**

Judgement about a tool is built slowly and specifically: where this model is strong, how it fails, which jobs are worth handing it and which waste a morning. It is hard-won and it is not portable, so change the tool and most of it goes. A team moved three times in eighteen months is permanently near the beginning, and a team near the beginning has no basis on which to challenge anything, because it does not yet know what good looks like.

Now put a person inside that. They are expected to produce at a rate the tooling is supposed to make effortless, and told that judgement is the valuable part of their job, while a policy specifies which models, which cases and which patterns are permitted. The boundaries were drawn before anyone had the chance to find out where the useful ones are.

So what is the rational move? Not to argue with the output. Arguing is slow, invisible to every measure anyone is applying, and produces a worse-looking week than accepting. Taking the first answer is fast, it meets the number, and it is defensible because it came from the approved tool used the approved way. Challenging is expensive, unrewarded and, under a prescriptive enough policy, mildly transgressive.

Which is where Lee's finding joins up. **People push back when they believe their own judgement is worth something.** An organisation that talks up what the tool can do, while narrowing the room people have to build and prove their own judgement, squeezes both ends at once.

I built one of these myself.

When I first deployed AI into my team I said the right things. I talked about ownership. I was clear that they stayed in control and stayed responsible for the output. The poster on the wall had the right headlines. What I measured was velocity: tickets completed, per person, per sprint.

Nobody had to be told which of those counted. One was a value and the other was a number, and only the number appeared in a review. It also moved in one direction. There was nothing in it for rejecting a suggestion, for deciding the tool was the wrong instrument for a job, or for spending a morning finding out where it fell over. On the measure I had chosen, all of those look identical to being slow.

The training pointed the same way. The workshops taught a fixed set of skills and prompts, and we built agents to take the thinking off people's hands. Useful, and I would do parts of it again. But none of it is agency. I was teaching operation and calling it capability, which is section 1's problem arriving in the one place I controlled completely.

What I should have been doing is harder to put on a slide: showing people where their experience was the thing the model did not have, and where the balance sat between going faster and being right.

I cannot prove the metric changed anybody's behaviour. What I can say is that I asked for a number that could only improve by handing over more, attached nothing at all to the judgement side of it, and was then surprised when judgement turned out to be what we were short of.

Autopilot is not a character flaw. It is the behaviour the system is paying for. If accepting is fast and rewarded, and challenging is slow and invisible, you will get acceptance, and you will get it from exactly the conscientious people you were relying on to notice things.

The organisation cannot replace what it stops producing

So far this is a problem about individuals, and as problems go it is manageable. Skill fades slowly and it comes back. Put someone on hard problems unaided for a few months and most of what they lost returns.

The organisational version has neither property. **It is not slow, and it does not come back.**

Junior roles were built out of precisely the work that is now cheapest to automate: the tedious implementation, the first-draft test suite, the small bug fix, the documentation nobody wanted to write. That was never only output. It was the apprenticeship. Automate all of it and you have not made a staffing decision. You have capped how much judgement the organisation will be able to apply to anything, a decade from now, without ever framing it as a decision about judgement at all.

Whether this is already showing up in hiring is contested. One study of US payroll records finds employment falling sharply among workers aged 22 to 25 in the occupations most exposed to AI (Brynjolfsson, Chandar and Chen, 2025). A review of the field argues that such claims are premature, and that the declines track interest rates at least as well as they track AI (Kolko, 2026). I am not going to settle that here and the argument does not need me to.

What is not contested is that organisations are restructuring now, on the assumption that it is settled. That is the failure section 11 describes: a decision that is expensive to reverse, taken on a premise nobody checked.

How people learn when the approved route disappears

The mechanism matters more than the headcount, and it holds even if hiring never moves at all.

Something close to this has already happened to surgeons, and it was studied properly while it happened.

In open surgery the trainee stands at the table. They hold, they cut, and they are handed more of the operation as they show they can take it. Nobody schedules that learning; it happens because they are physically in the work, next to someone better than them. Robotic surgery moved the lead surgeon to a console a few feet away, and one person now carries out the operation through the machine. The trainee watches a screen.

Matthew Beane spent two years inside that shift, with fieldwork at five hospitals and then a blinded interview study across thirteen of the top US teaching hospitals. The approved way of learning the job had quietly stopped working. **Only a minority of trainees reached competence.**

The ones who did get there did it by breaking the rules. Beane calls this **shadow learning**: specialising earlier than permitted, rehearsing abstractly, and above all seeking out what he terms *undersupervised struggle*, operating at the edge of their capability without the supervision that policy required of them (Beane, 2019). This is surgery rather than software, and the transfer to engineering is an argument I am making rather than a finding he reports.

The apprenticeship was never the tasks, it was the *participation*: being close enough to the real work, often enough, to acquire judgement nobody could simply tell you. Automate the tasks and you have not removed some output, you have removed the route.

The second consequence is worse. **The people who learn anyway will do it by breaking your rules.** That is not a prediction, it is Beane's finding, and it lands directly on section 1. An organisation that automates the entry-level work *and* prescribes exactly how AI may be used has closed the sanctioned route and the unsanctioned one at once. What it gets is not compliance. It gets the same shadow learning, driven further out of sight, which is precisely the pattern section 7 documents happening now.

The mechanism that worries me most is not a decision at all, and it does not require the hiring argument above to be settled. Falling entry-level recruitment has been reported widely enough that the trend needs no introduction, though section 11 is precisely about what happens when organisations plan against headlines. The version that needs no headline is attrition. People leave, they are not replaced, and the junior intake thins without anyone visibly choosing it: no announcement, no debate, just a graduate scheme that is not run this year and then not run the next. **That is agency loss in its purest form: a consequential, hard-to-reverse choice, made by default, by nobody.**

Section 5 described judgement being exercised less. This section describes the supply of it being cut off, individually as practice decays, and structurally as the route by which people acquired it is automated away and then policed. **Accountability, meanwhile, has not moved an inch.** The

organisation will still be answerable, in ten years, for decisions it will have fewer and fewer people qualified to make.

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7. What people do when they feel agency slipping

People who feel their control over their own work is slipping do not generally say so. They adapt, privately.

The best-evidenced version of this is the shadow use data. Across 48,340 people in 47 countries, more than half of employees conceal their AI use and present its output as their own, and **48–49% have uploaded sensitive company or customer data into public tools** (Gillespie et al., 2025).

Most telling for anyone writing policy is what happens as the controls tighten. Where there was **no policy at all, inappropriate use ran at 33%**. Where a policy existed, it rose to **56%**. Where AI had been banned outright, it reached **67%** (Gillespie et al., 2025).

Read that middle number again, because it is the one that applies to almost everyone reading this. Very few organisations have banned AI, and if yours had, you would know. What most people are working under is a policy: an approved tool, an approved list of use cases, a set of things you must not put into it, and a process for asking permission to do something else. That is the normal, reasonable, well-intentioned middle. **And it is associated with inappropriate use running at nearly twice the rate of having no policy at all.**

The causality may well run the other way, since organisations often write a policy after they have found a problem. But what the gradient does rule out is the assumption that tighter control produces better behaviour. On this evidence it does not, at any point on the scale.

The agency reading is the one that matters here. A policy written early, before anyone in the organisation has enough experience to know what the rules should say, is not really governing practice. It is guessing at practice and then freezing the guess. People hit the edges of it within a fortnight, find that the approved route will not do the thing they need, and quietly go around. The behaviour does not stop. It relocates, onto a personal account, outside anything the organisation can see.

What tight policy buys is not control. It is blindness. The same work carries on, now unobservable, unreviewable and ungoverned, and the organisation has swapped real agency for the appearance of it. The posture looks responsible right up until the moment someone asks what is actually happening, and nobody can answer.

This is also where section 4 comes back. Telling people that judgement is now their contribution, while prescribing the judgement in advance, does not produce better judgement. It produces people who have learned that the honest answer to "how should I use this?" is one they are better off not giving out loud. The policy has not raised the standard of practice. It has hidden it.

And it closes the loop opened in section 1. The rules arrive early, before anyone has enough experience to know what the rules should say. They remove the licence to explore, so the learning does not happen in the open. It happens anyway, privately, with nothing fed back into what the organisation knows. Every protection the policy was written to provide is defeated by the policy.

There is a further cost, and it does not stay confined to AI.

Ron Westrum spent his career studying how information moves through organisations in high-risk settings such as aviation and medicine. His argument is that you can read a culture directly from its information flow, because "*information flow is both influential and also indicative of other aspects of culture*" (Westrum, 2004). He sorts organisations into three types: pathological, which is power-oriented; bureaucratic, which is rule-oriented; and generative, which is performance-oriented.

Look at what sits in the rule-oriented column. Messengers are neglected. Responsibilities are narrow. Bridging between teams is tolerated rather than encouraged. And then the row that should stop you: **novelty leads to problems.**

Not novelty *crushed*, which belongs to the pathological column and at least has the merit of being visible. Novelty leads to problems. The new idea is not forbidden. It is merely inconvenient, and everyone works out quickly that having one generates exceptions to request, approvals to chase and awkward conversations to sit through. So fewer people have them. The ones who do keep them quiet.

That is the drift we risk inviting when the response to something we do not yet understand is to wrap it in tighter rules. The instinct is entirely reasonable and I have had it myself. But in Westrum's terms it is a step from generative towards bureaucratic, and the uncomfortable part of his argument is that the step does not stay confined to the thing we were worried about. It is a disposition rather than a rule, and dispositions travel. Which makes it worth being conscious of while the rules are being written, rather than a verdict on anyone who has written some.

DORA has measured what that costs. Its research finds that "*organizational culture that is high-trust and emphasizes information flow is predictive of software delivery performance and organizational performance in technology*" (DORA, n.d.).

Which lands awkwardly next to section 3. AI adoption is already associated with falling delivery stability, and the natural response to falling stability is more control. But more control degrades the one cultural variable DORA found predicts delivery performance in the first place. The remedy pushes against the outcome it is reaching for.

You can own licences for the fastest tooling on the market and still lose, because nobody has been left any room to have an idea. The speed at which code can be generated was never the binding constraint. The constraint is how many good ideas reach the surface, and how quickly the organisation finds out it was wrong about something. Both of those are information flow, and both are what tight policy quietly taxes.

None of which is an argument against having a policy at all, and I want to be careful not to be read as making one. It is an argument against writing it first, writing it tightly, and leaving it unchanged while

everybody's understanding moves underneath it. Section 13 sets out what the alternative looks like in practice.

Amy Edmondson's mechanism is the relevant one for the alternative: psychological safety improves performance because it enables **learning** under uncertainty, not because it makes people comfortable (Edmondson, 1999).

So far this has read as four separate problems. Rules that foreclose learning. Judgement that fades through disuse. An apprenticeship that no longer happens. Practice that goes underground. They arrive in different places and they feel unrelated.

They are not. Look at what each one does to information.

Concealed use means the organisation cannot see how the work is actually being done. Unexercised judgement means the reasoning behind a decision is never articulated, so there is nothing to record even if somebody wanted to. A missing apprenticeship means the people who would have held that reasoning in ten years' time are not being made. Prescriptive policy means the honest account of what works is the one thing nobody can safely give.

Every mechanism in Part I removes information from the organisation. That is the single thing they have in common, and Westrum is the reason it matters: an organisation substantially *is* its information flow. Starve that and you have not inconvenienced the organisation. You have changed what kind of organisation it is.

Which is the hinge of this paper, and it is worth stating plainly. **Organisational agency is not a separate thing that gets lost separately. It is made out of what individuals export, and Part II is the consequence of Part I rather than a second story running alongside it.** Nobody has to decide to give up organisational agency. It drains, through a hundred reasonable individual choices, none of which looked like a decision at the time.

It also behaves worse than the individual version. A person who stops exercising judgement can get it back; put them on hard problems unaided and most of it returns within months. An organisation cannot recover reasoning that was never written down. There is nothing to restore it from. The knowledge did not move somewhere less convenient. It stopped existing.

The rest of this paper follows what that costs, and it starts with the quietest version. Before an organisation loses the ability to explain its decisions, it loses the ability to notice that a decision was taken at all.

PART II — HOW ORGANISATIONAL AGENCY FOLLOWS

8. Ambiguity is a decision you have delegated without noticing

When intent is unclear, AI does not stop. It fills the gap — sensibly, confidently, using whatever context it has and whatever patterns it learned in training. Often the assumption is perfectly reasonable. Reasonable and correct are not the same thing.

In the old world, ambiguity created friction. A developer paused, asked a question, waited for an answer, or worked out halfway through the implementation that the ticket made no sense. Irritating, certainly. But that friction was a checkpoint doing useful work — and, read correctly, it was a **mechanism for returning decisions to the people who should have been making them.**

AI removes it.

An agent makes an assumption and carries on within seconds. That assumption shapes the next decision, which shapes the one after. A vague requirement becomes a design choice, the design choice becomes implementation, and then tests are generated around the implementation to confirm that everything works exactly as intended.

The result is software that is internally consistent, well tested, and nobody chose.

This is not a hypothetical. Orchid is a benchmark of 1,304 function-level tasks built deliberately around four kinds of ambiguity: lexical, syntactic, semantic and vagueness. Across every model tested, the authors report that "*ambiguity consistently degrades the performance of all evaluated LLMs, with the most pronounced negative effects observed in highly advanced models*" (Yang et al., 2026).

The next finding is the one that belongs in this section. The same authors observe that models "*frequently produce functionally divergent implementations for the same ambiguous requirement and lack the capability to identify or resolve such ambiguity autonomously*". Read that twice. One requirement, handed over more than once, produces software that behaves differently each time. Which version you end up shipping is not something anybody decided. It is closer to a coin toss conducted on your behalf, and the losing side of it still arrives with passing tests.

Orchid is a preprint, and its tasks are single functions rather than features, which is a fair thing to hold against it. It cuts the other way, though. A single function is the smallest unit of ambiguity there is, so a model that cannot hold intent steady across one of those will not do better across a feature, a service, or a week of agent work.

Two things follow from that, and the second is the one people miss.

First, **ambiguity gets more expensive as capability rises**, not less. A weaker model fails visibly and you catch it. A stronger model produces something coherent and defensible built on an assumption you never saw.

Second, **you cannot delegate disambiguation to the machine.** The decision about what the product should do when the specification is silent is a product decision. If it is not made by a person, it is made by a training distribution.

And the compounding is the real danger. As we hand over bigger tasks, chain agents together and let them run unsupervised for longer, every decision becomes the context for the next one, so a small deviation at the start drifts quietly until the finished thing is a long way from where anyone meant to be. Model success rates fall as task length rises (METR, 2026). And when researchers annotated 1,642 execution traces across seven multi-agent frameworks, weak verification of the result turned out to be a failure category in its own right. Adding a step that checked the work against the high-level objective improved success by 15.6% (Cemri et al., 2025). Both are preprints, and Cemri's frameworks are research systems rather than production ones.

And this is where agency actually goes. Not dramatically, and not by anybody's decision. The output arrives, it looks right, and it is accepted. That is convenience meeting the need to perform, and section 6 explains why both are pulling in the same direction.

But notice what gets skipped. The output was never the first place to apply judgement. **The largest influence anyone has over what comes back is what goes in**: the context they supply, and the context they understand well enough to supply in the first place. Someone who knows the system, the constraint nobody wrote down, the reason the last attempt failed, is holding the one thing the model cannot obtain for itself. Spending that at the start is worth more than catching the consequences of withholding it at the end.

That is the education we have not provided, and the gap is tangible. We have taught people to operate the tool, and some of us have taught them to check its work. We have largely not taught them that the quality of the answer is mostly settled before anyone presses enter.

Agency does not have to disappear. But it does have to move. It moves from producing the output to framing the problem the output is answering: what is actually being asked, which constraints are real, what must not be assumed, and what would have to be true for the result to be acceptable. That is not a smaller job than the one it replaces. It is a harder one, and it is the part that stays human for considerably longer.

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9. Knowledge that never leaves the chat window

An organisation's agency rests on knowing why its own systems are the way they are. That knowledge has always been fragile. AI changes the economics of the fragility.

Documentation has always been the first thing to slip and the last thing to be valued. It was survivable, because the knowledge lived in people; you asked the person who built it. That trade-off no longer holds, for the obvious reason that **AI cannot ask the colleague who left last year**, and for a less obvious one: increasingly, there was no colleague. The reasoning existed for ninety seconds inside a context window and was discarded when the tab closed.

So undocumented behaviour has gone from inconvenient to actively expensive. The team that cannot explain its own system cannot direct the tool that now builds it.

The pattern I would warn against most strongly is the one that looks most like productivity: **one person and their agent**, moving extremely fast, with every piece of context assembled on their machine and nothing durable left behind. The output ships. The understanding does not exist anywhere an organisation can reach.

Type	Lifespan	Examples	What losing it costs you
Task	Hours	The specific change, the ticket, the file	Rework. Cheap.
Project / initiative	Weeks–months	Scope, decisions, constraints, domain model	A team that cannot explain its own initiative
Product	Years	How the product behaves, why it works that way	Change becomes archaeology
Organisational / strategic	Long-lived	Standards, architecture principles, security posture, compliance	Every initiative re-decides what was already decided

Most organisations manage the first tier well and the rest badly. The higher tiers are the reusable ones, which is exactly why neglecting them compounds.

The evidence on this is better than people expect. DORA finds documentation quality acts as a **multiplier on every technical capability studied**, and a substantially larger organisational-performance effect for teams with above-average documentation (DORA, 2023).

And there is a pleasing irony in the 2024 finding: AI measurably **improved documentation quality** while delivery stability worsened in the same data (DORA, 2024). The capability to fix this is in the box. It is the deliberateness that is missing.

Context debt accrues interest exactly like technical debt, and it is less visible, because nothing breaks.

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10. You do not control the model. You control the harness

This is the section most likely to be new to readers, and it is where organisational agency is most often conceded without anyone noticing.

A **harness** is everything built around a model to direct it: the templates it fills in, the stages it must move through, the artefacts it must produce before continuing, the checks that run on the result. The model supplies capability. The harness decides what that capability is pointed at and what counts as done.

Same model, different harness, markedly different output. Which is why the endless argument about which model is best is mostly beside the point for a delivery organisation.

You will not improve the model. Everything around it is yours — and that is where outcomes diverge.

Three consequences for agency.

The harness is the part you control. It is also the part most organisations outsource by default, by adopting a framework wholesale and never examining it.

A harness encodes an opinion. Every gate in it is somebody's view of what goes wrong when you skip that step. Adopting one uncritically means inheriting a set of judgements about your own risk profile, made by people who have never seen your system. That may be fine. It should at least be known.

Harnesses are disposable; your specifications are not. Tooling in this space has a shelf life measured in quarters. The record of what your systems do and why has to outlive it. If your knowledge is only legible inside one vendor's framework, you have made an exit expensive on purpose.

And there is a failure mode specific to harnesses that is worth naming, because it looks exactly like success: green stages, complete templates, a full audit trail, and a specification no human has read closely enough to disagree with. The tool is satisfied. Nobody is any clearer.

Behind all of this sits a harder constraint: **the thing in the middle is not stable.**

Model behaviour changes between versions in ways the vendors themselves do not anticipate. In April 2026 OpenAI published a post-mortem explaining why ChatGPT had started talking about goblins. A reward signal attached to one personality had favoured creature metaphors, and the tic then generalised well beyond that setting through reuse of training rollouts. On OpenAI's own figures it reached 0.12% of conversations, up from 0.04% (OpenAI, 2026).

That is a small number, and its size is not the point. **Nobody intended it, nobody predicted it, and the vendor needed a forensic audit to work out why it had happened.** The GPT-4o sycophancy incident makes the same case more seriously: an update that made the model markedly more flattering and agreeable than intended, shipped to production and publicly rolled back (OpenAI, 2025).

Add to that the separation Narayanan and Kapoor draw between capability and reliability: "*If an AI system performs a task correctly 90% of the time, we can say that it is capable of performing the task but*

it cannot do so reliably. The techniques that get us to 90% are unlikely to get us to 100%." (Narayanan and Kapoor, 2024)

Put together: behaviour shifts in ways vendors do not anticipate, capability is outrunning reliability, and we are handing over more complete control at the same time. That is not an argument against using AI. It is an argument for owning the structure around it.

And it is where the fourth part of agency finally comes due. Authorship, judgement and control have all moved in this paper. Accountability has not moved an inch. What has thinned is the ability to discharge it. You remain answerable for decisions you did not personally make, arrived at through reasoning that was never recorded, produced by a system that may not behave next month the way it behaved when you approved it.

Each of those is survivable alone. Together they take away the thing accountability actually requires, which is not authorship but the ability to reconstruct: what was intended, what was assumed, what was rejected and why. That is the minimum viable form of organisational agency, and very nearly everything else is recoverable from it.

Which puts a name to the governance mistake worth avoiding. The instinct, when this is pointed out, is to strengthen the gate. More approvals, more sign-off, more evidence. Some of that is sensible. But the failure mode is **automating the approval rather than supporting the judgement**, and a faster gate that encodes no judgement is worse than the slow one it replaced, because it manufactures evidence of diligence that did not occur. The audit trail gets better while the thing it is evidence of gets worse.

I should be straight about the status of this. I know of no published incident that has turned on an organisation being unable to explain AI-generated code, and I would not cite one if I did, because the case does not need it. It follows from things this paper has already shown: reasoning that lives in a context window and is discarded (section 9), and a model underneath that changes without asking (this one). The bill for that arrives late, which is precisely why it is easy to defer.

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11. Decisions taken on an unverified premise

Organisational agency also fails in a plainer way than anything in the last three sections. An organisation makes a structural, hard-to-reverse decision on the basis of a capability claim that nobody checked.

The claim usually arrives with a number on it, and the most durable of them is 10x. Nothing supports it. Credible measured uplift clusters around **20–30%**, across a range running from a 55.8% speedup on a toy greenfield task to a **19% slowdown** among experienced developers working in mature codebases they knew well (Synthesis of controlled studies, 2023–2026). Those studies measure different outcomes and are not directly commensurable, so the spread is the finding rather than the midpoint. **No controlled study supports anything approaching 10x.**

Decisions were taken on it anyway. The most quoted is Klarna, and the popular version is **false on both halves**. The company did not fire 700 people and then rehire them. 700 was a workload-equivalence estimate, never a redundancy count, and headcount was still falling at IPO (Klarna Group plc, 2025; Klarna, 2024; CNBC, 2025).

The real story is more useful than the myth. Klarna admitted *quality* degradation, moved to an explicit "**dual-track approach**" pairing AI with high-quality human support, and quietly restated a "$40M profit" claim as "$39M cost savings" (CNBC, 2025; TechCrunch, 2025; Klarna Group plc, 2025). That is a role redesign, arrived at the expensive way.

Amazon shows the same thing from the other end: cuts routinely attributed to AI which the CEO explicitly denied were AI-driven (GeekWire, 2025).

"AI" has become a socially acceptable label for decisions taken for other reasons, and that corrupts the evidence base everyone else is reasoning from.

There is a growing "AI boomerang" genre: companies that cut staff for AI, regretted it, and hired them back. Every one of those stories supports the argument I am making here, which is precisely why I should not take any of them on trust. Most do not survive contact with the source.

The IBM version has it that the company sacked 8,000 people to AI and quietly rehired them. What was actually said, in 2023, was that IBM expected to **pause hiring** for back-office roles, and that perhaps 30% of around 26,000 such roles might be replaced by AI and automation over five years (Bloomberg, 2023). Thirty per cent of 26,000 is roughly 7,800, which is where the 8,000 came from. Nobody was sacked, so nobody was rehired. IBM later put the actual displacement in the low hundreds, and total headcount rose.

The survey everyone quotes deserves more care still. Orgvue is reported as finding that **55% of businesses regret making people redundant because of AI**. The actual finding is that 39% of leaders had made AI-related redundancies, and 55% *of that group* thought they had got some of those decisions wrong, which is around **21% of businesses, not 55%** (Orgvue, 2025). It is also vendor-commissioned, since Orgvue sells workforce-planning software, and it measures self-reported regret rather than rehiring, which is an inference the press added afterwards.

Every one of these is the same mechanism. A figure is detached from its base, the qualifier is dropped in transit, and it arrives somewhere as a settled fact that somebody then plans against.

Which is the agency point. An organisation that restructures on a claim it has not verified has not exercised judgement. It has outsourced a strategic decision to a press cycle. That is section 5's failure at organisational scale, and it is the more embarrassing version, because the evidence was public, the arithmetic was trivial, and nobody spent the afternoon it would have taken to check.

PART III — AGENCY IS A DESIGN CHOICE

12. Choose the altitude of delegation deliberately

Everything so far has been diagnosis. This part is about what you do with it, and it begins with the decision the whole paper has been circling: not whether to delegate, but how far.

Section 3 left an aircraft on autopilot with nobody watching the instruments. The part of that comparison I did not use then is the part that matters now. Autopilot is not one switch. A crew selects a mode, and the modes differ in how much they hand over, from holding a heading to flying an entire approach. They choose it, they say out loud which one is engaged, and they know what remains theirs in each. Nobody treats the most automated mode as the best one.

Delegating to AI works the same way, and is almost never treated that way. The question is rarely "*should we let AI do this?*" It is "*at what altitude did we hand this over, and did anyone actually choose it?*" In most organisations the second half of that has no answer. There is one setting, applied to everything, arrived at by accident and never said aloud.

Two things should decide it, and neither is the one teams tend to use. **Blast radius** is how much breaks if this goes wrong: one screen, one service, or every customer. **Reversibility** is how easily you can undo it once you notice. Together they tell you how much you can afford to hand over. How large or how technical the job is tells you very little.

What follows is five settings, not five stages.

Level	What the human does	What the machine does	Entitled to use it when	Where your agency lives
0 — Manual	Everything	Nothing	Novel, high-consequence, no precedent	Everywhere, by default
1 — Assisted	Authors; accepts or rejects suggestions	Completes, suggests	You can evaluate a suggestion as fast as it arrives	The rejection
2 — Delegated task	Specifies precisely; reviews output	Implements a defined change	You have acceptance criteria you could hand to a stranger	The specification
3 — Delegated objective	States intent and constraints; reviews result	Breaks down the work, chooses the approach, implements	Blast radius is contained and an external check exists	The constraints, and the check
4 — Autonomous	Sets the goal; samples outcomes	Everything, including decomposition	Reversible, low-consequence, and you can see what it did	Deciding what qualifies, and sampling often enough to notice when it stops qualifying

Every level on that table is legitimate. This is not a maturity model and 4 is not the destination. It is a matching exercise, and treating it as a ladder to climb is itself the agency failure this paper is about. The level belongs to the piece of work, not to the team or the tool, with the repository supplying a default rather than a rule. Most organisations currently run one setting for everything, arrived at by accident.

Read the last column downwards and the real pattern appears. **Your contribution moves earlier the more you hand over.** At level 1 you judge the output. By level 3 there is no meaningful moment to judge the output, because the approach was chosen without you, so your judgement has to have been spent already, on the constraints and on what the check tests for. This is section 8's point arriving as a mechanism rather than a slogan. Agency does not disappear as you delegate more. It relocates to the front of the work, and if you have not moved it there deliberately, it is simply gone.

Which is why each level has its own characteristic failure, and all of them are in this paper already. Level 1 fails as section 5 describes: accepting becomes faster than evaluating, and the rejection you were supposed to be contributing stops happening. Level 2 fails as section 8 describes: the specification is vague, so the decision is delegated along with the task and nobody notices. Level 3 fails as sections 9 and 10 describe: you see a result and never the reasoning, so there is nothing to reconstruct later. Level 4 fails through drift, where the thing that genuinely qualified as reversible and low-consequence quietly stops being either.

The requirement at level 3 is the one people skip, and it has the firmest evidence behind it. Models cannot reliably correct their own reasoning without an external signal, but improve substantially when an external check gates the loop (Huang et al., 2024), and in planning tasks an external verifier lifted performance from roughly 35% to 82% (Kambhampati et al., 2024). The check has to sit outside the model. A model asked to mark its own homework is not a check. Separately, Anthropic's usage data

reproduces METR's finding that success rates fall as tasks run longer (Anthropic, 2026a; METR, 2026), which is the argument for handing over less the longer something runs unattended.

Drift is upward and silent, because no one ever decides to move from 2 to 3. It happens one convenient afternoon at a time. The fix is not a committee, and it does not need to be heavier than a line of text: state the level in the pull request, next to the work it describes. That is enough to make it sayable out loud, and most of the value is in the saying. A team that cannot state which level it is on has usually handed over more than it thinks, and the moment to re-ask the question is when blast radius changes, not when a calendar says so.

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13. People first, not tool first

Everything in the last section was about where to put the work. This one is about who is still in it, and it is the part most rollouts skip entirely.

A tool-first rollout makes an assumption it never states: that the value is in the tool, so the job is distribution. Procure it, licence it, publish the rules, measure the uptake. It is efficient, it is legible to a board, and it is what I did. The assumption is wrong in a specific way. The tool is now the cheap part and it is getting cheaper, which means it is also the part your competitors have. What is scarce is people who can tell whether what came back is any good, and who will say so when it is not.

Put that way, the investment looks obviously mis-sized. We spend significant money on licences and almost nothing on the thing that determines what those licences produce.

A people-first approach starts from a harder question: when the machine does the heavy lifting, what exactly is a person for? Section 4 argued that the honest answer is directing, judging and owning rather than producing. That is a decent summary and far too abstract to act on. So it is worth being concrete about the behaviours actually worth nurturing, because they are specific, they are learnable, and almost none of them appear in standard AI training.

- **Saying what you want before asking for it.** The most consequential skill in the list, and the least glamorous. Section 8 is the evidence: the largest influence anyone has over the output is what goes in.
- **Noticing when a decision is being taken.** The ability to look at a thin ticket and see the four unanswered questions inside it, before a model answers them fluently and invisibly.
- **Specific scepticism rather than general scepticism.** "Always check AI output" is advice nobody can act on. Knowing the three ways a model characteristically fails in *your* domain is a working skill, and it takes weeks of exposure to acquire.
- **Being able to say why something is wrong, not just that it is.** The second is a reflex. The first is judgement, and it is the only one of the two that improves anything.
- **Writing down the reasoning and not only the result.** Section 9's remedy, in the form of a habit rather than a tool.

- **Knowing when to stop delegating.** Looking at a piece of work and deciding how much of it to hand over, based on what happens if it is wrong and how hard it would be to undo. That is a judgement about consequence, not about what the model is capable of, and section 12 is the whole of it.
- **Disagreeing with a fluent answer while the clock is running.** This one is organisational as much as personal. Sections 5 and 6 explain why it is the first to go, and no amount of training will produce it in a system that rewards acceptance.

That last point is the difference between this and a competency framework. These behaviours are not a syllabus to be delivered. Most of them cannot be taught in a room at all, because they are built out of contact with real work. Which raises the question underneath this entire part of the paper.

Where the judgement comes from

Every behaviour on that list assumes somebody has the judgement to exercise it. Section 6 is the problem with that. The route people used to take to acquire judgement, being close enough to real work often enough that it accumulated without anyone scheduling it, is the route we are currently automating. So the question underneath all of this is not which behaviours to name. It is where the next lot of judgement is going to come from.

It is not going to come from a course. Section 6 already named that trap, in my own hands: teaching the operation of a tool and calling it capability. A fixed set of skills and approved prompts produces people who can work the machine, which is a different thing from people who can tell when it is wrong.

What seems to build it is cheaper and less tidy than a curriculum. Let people experiment against the real work they are responsible for: run the same piece three different ways, build a crude workflow of their own before adopting anybody else's, and look properly at what came back each time. **The point of building one is not the artefact.** Over a few weeks something specific forms, in two places at once. One is an opinion about what good and bad output look like in *their* domain, which is never the generic version in the training material. The other is the ability to read somebody else's harness: having decided for yourself which steps were worth enforcing, and found out which ones you skipped and regretted, you can pick up Spec Kit or Kiro or AI-DLC and see what each gate is for and whether the problem it solves is one you have. Both of those are judgement, and neither can be handed over, because both are made out of having been wrong a few times in a context that mattered. Adopt first and you get neither, because you have nothing to compare against and the thing arrives looking authoritative.

The loop is the point, and it is worth stating plainly: change the input, change the workflow, look at what actually changed in the output, and decide whether that is better. Deciding whether it is better is the whole thing. It is not measurable from outside, it does not show up in a sprint report, and it is the one part no model can do on your behalf, because "better" is a claim about what you were trying to achieve. This is also where section 8 lands: if the biggest influence on the output is what goes

in, then the skill worth building is knowing what to put in, and that is learned by watching what different inputs do.

Two things have to be true for any of this to happen. People need time that does not count against delivery, which means somebody senior has to decide that the velocity number is not the only thing being watched. And the choosing has to be real. Section 10 made the point that a harness encodes somebody's opinion about what goes wrong when you skip a step, which may suit you and may not. The only reliable protection against inheriting policy you did not write is having written some yourself, and then being free to pick. An organisation that distributes one approved workflow and calls it enablement has bought the artefact and removed both halves of that.

None of this is far from Beane. The trainees who got good sought out undersupervised struggle, and did it against the rules. The difference available to us is that we can simply allow it, rather than making people take it without permission. The organisation that will not make room for it does not avoid it. It gets the same thing, done quietly, by the people who were going to learn anyway, and loses the only useful part: what they found out never comes back.

What good looks like, and where the behaviours go

Nurturing behaviour is famously soft work. It resists measurement, it is easy to fake with a values poster, and most attempts at it produce a competency framework nobody reads. So it is worth having something concrete to aim at, and section 7 already introduced it from the other side.

Westrum's typology is usually read as a diagnosis, which is how I used it earlier: the rule-oriented column, where messengers are neglected, responsibilities stay narrow, bridging between teams is tolerated at best, and novelty leads to problems. The performance-oriented column is the half I did not use. There, somebody goes looking for information before they are handed it. Messengers get trained, because bad news arriving early is worth paying for. Responsibilities cross team boundaries instead of stopping at them. A failure prompts the question of what happened, not the question of whose fault it was. And a new idea gets built (Westrum, 2004, as reproduced by DORA, n.d.). Worth saying plainly that this is a typology drawn from aviation and medicine, not a measured finding, and that the correlation with delivery performance comes from vendor-published survey data. It organises what to look for. It does not prove the direction of the arrow.

Read the seven behaviours against that column and they stop looking like personal virtues. Every one of them is a behaviour an organisation either rewards or quietly taxes. Specific scepticism requires novelty to be implemented, because you cannot learn how a model fails in your domain without trying things that might not work. Writing down reasoning is information being actively sought. Noticing that a decision is being taken, and saying so, is the messenger behaviour, and section 6 is an account of what happens when that is expensive.

Which reframes the whole exercise. You are not trying to produce better individuals. You are trying to be the kind of organisation where those behaviours are the path of least resistance rather than acts of personal courage. That is tangible enough to act on, because it gives you a

question to ask of any change you are considering: does this make it more likely or less likely that somebody tells you something you did not want to hear?

One honest limitation, which is really the bridge to what follows. Behaviours nurtured in individuals stay in individuals. They leave when people leave, they do not transfer between teams, and they are invisible to anyone not in the room, which is section 9's problem arriving from a different direction. A person with excellent judgement and nowhere to put it is still a single point of failure.

So the behaviours need a channel: a shared way of working that takes what people have learned and makes it visible, reviewable and inheritable. That is what the next section is. Spec-Driven Development is the name for how the specific moves hold together. **The order matters, and getting it the wrong way round is the original mistake repeating itself.** Behaviours first, then the discipline that carries them. A practice adopted by people who have not built the judgement it assumes is a harness with nobody at the controls.

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14. Spec-Driven Development, and where the judgement goes in it

I have used the name once already without explaining it, so let me do that before arguing for it.

Spec-Driven Development is the practice of making intent explicit before execution, and keeping it current as understanding changes. Specifications are obviously not new. We have had requirements, architecture documents, user stories and acceptance criteria for decades, and we have been quietly letting them go stale for just as long. That was survivable, because a specification was mostly people communicating with people and the code was the artefact that actually mattered. If the document drifted, the team still knew.

AI changes that arithmetic. The specification stops being a description of the work and becomes the thing the work is generated from: the shared account of intent that both people and models reason out of. A stale specification used to be untidy. Now it is an instruction. In practice SDD is a short list of habits, and none of them are exotic: understand the problem, the outcome and the boundaries before asking for anything to be built; write down the assumptions and the open questions rather than letting them be settled in the middle of an implementation; describe the behaviour you need before the mechanism; say what evidence would prove the thing works; keep the record next to the code and change it in the same pull request.

It is worth separating the discipline from the tooling, because this is the distinction people collapse most often. GitHub's Spec Kit, Kiro, AWS's AI-DLC and the rest are harnesses: scaffolding built around a model that decides which stages it must move through, which artefacts it must produce, and what counts as done. The model supplies capability. The harness decides what that capability is pointed at. SDD is not the harness. SDD is the discipline a harness is one way of enforcing, and the two come apart in both directions. You can practise it with a markdown file and the habit of writing things down first. You can also run a harness end to end, produce every artefact it asks for, and be doing nothing

recognisable as SDD, because the documents were written to satisfy the tool rather than to capture what anybody intended.

The harness is also a way to lose it

Which is worth dwelling on, because a harness is not a neutral container for the discipline. It is a second place agency goes, and the loss there is subtler than anywhere else in this paper, because the whole apparatus was built to make people think harder.

Two things happen. The first is that the input bends to the tool. You stop asking what this piece of work actually needs and start asking what the template wants, which is a smaller and much easier question. Fields get filled because they exist. The consideration that had no box quietly does not get written down, and the constraint that mattered most is the one the form had no room for. Section 8 said the largest influence on the output is the input. A harness shapes the input, which means it is exercising a share of your agency on your behalf, usefully or otherwise.

The second is that structure reads as rigour. Output that arrives through four approved stages, with every artefact complete and every gate green, carries an authority the content has not earned. Completeness is not correctness. This is section 5 arriving by a different route: the review becomes a check that the process ran rather than a judgement about whether the thing is right, and it is harder to resist here because the evidence of diligence is so abundant.

Both get worse when the harness was handed to you. **Every gate in a harness is somebody's opinion about what goes wrong if you skip it**, and adopting one without knowing which opinions you have inherited is section 4's failure in tooling form: prescribed judgement, accepted as infrastructure. A harness you have argued with is a set of positions you hold. A harness you were issued and never touched is a set of answers to questions you never asked.

None of which is an argument against harnesses, or for every team maintaining one from scratch. Section 13's point was that you build one in order to be able to read the others, not in order to keep it. What matters afterwards is not who authored the one you end up on, but whether the team chose it and can still change it, because ownership shows up as modification rather than adoption. So the test I would apply is whether a team can say why a particular gate is there, and whether they have ever removed one. If every stage survives because it came in the box, the tool is running the team.

Where the agency actually sits

Section 12 found that your contribution moves earlier as you delegate more. A specification is the instrument for moving it, which is the whole reason this belongs in a paper about agency rather than one about process. It is what closes the gap section 8 described, where an unclear requirement gets filled in by a model and nobody notices a decision was taken. **The distance between human intent and machine execution is the variable, and it matters more the more autonomous the machine becomes.** When generation was slow, a weak specification could be papered over with conversation and constant intervention. That does not survive an agent working unattended for an afternoon.

So it is worth being specific about which decisions stay with the person, because "keep humans in the loop" is exactly the kind of phrase that sounds like a safeguard and commits nobody to anything.

The decision	Why it cannot be delegated
What problem we are solving, and what we are not	There is no fact of the matter for a model to retrieve. This is a choice about what the organisation wants
Which ambiguities are real, and how they resolve	Section 8: models do not surface ambiguity, they resolve it silently and fluently (Yang et al., 2026)
The trade-offs, and which way they go	Cost, risk, time and quality are weighed against a context the model does not have and has not been told
What counts as done, and what evidence proves it	Section 12 required an external check. Defining what the check tests for is the check
Whether the outcome is acceptable	Section 5: this is the review act, and it is judgement rather than approval

What is left, implementation, is a large amount of work and increasingly good work. That is the trade this paper has been describing from the start. The point is not that less is handed over. It is that what is handed over is the execution rather than the decision, and that distinction only exists if someone wrote the decision down.

This is also where the judgement is applied rather than merely retained, and the practical form of it is unglamorous: review the specification, not only the diff. Reviewing a diff asks you to reconstruct intent from an implementation, which is precisely the reasoning section 9 showed is not written down anywhere. Reviewing the specification asks whether the thing described is the thing you want, which you are qualified to answer and which stays cheap to change. That is the 15.6% finding from section 8 in organisational form: checking the work against the objective, rather than checking the work against itself.

The practices that carry it

Everything above is an argument. This is what it looks like as a set of moves, gathered from across the paper so they can be read in one place. Four of them are SDD proper: 1, 2, 5 and 8 are very nearly the whole of it. The rest come from earlier sections and hold the same line by other means.

They are practices, not policy. Section 4 argued that prescribing the judgement is the failure, so a team handed this as a standard has reproduced the problem rather than solved it. Each one is a hypothesis to test, adapt and break, and the value is in the arguing, which is also where the judgement gets built.

#	Practice	The agency it preserves
1	**Make intent explicit before execution**	Product decisions stay with people, because there is no gap for the model to fill
2	**Surface assumptions as a reviewable artefact**	You see the decisions being made on your behalf, while they are still cheap to change
3	**Work in verifiable increments**	Course correction stays possible; compounding error gets interrupted (Cemri et al., 2025)
4	**Keep humans at the consequential decisions**	Judgement is spent where it matters, not spread thin across everything
5	**Review the specification, not only the diff**	Review stays an act of judgement. You check intent, which you hold, instead of reconstructing reasoning nobody wrote down
6	**Sample deeply rather than skim broadly**	Attention survives volume; the near-miss gets caught, which skimming never does
7	**Plan review capacity explicitly**	Judgement remains real rather than ceremonial (DORA, 2025)
8	**Specifications live in the repo, change in the same PR**	Organisational memory survives the people and the tooling
9	**Small batches, fast recovery**	Being wrong stays cheap — which is what makes delegation safe (DORA, 2025)
10	**Scale rigour to blast radius**	Control is spent where it buys something, not applied uniformly

Three honest qualifications

None of this is evidenced the way the earlier sections are. There is no controlled study showing that spec-driven teams retain more agency, and I am not going to imply otherwise. It is an argument from mechanism, and it should be read as one.

It has an obvious failure mode, and the harness subsection above is half of it. A specification written to satisfy a tool is not a specification, it is a receipt. The test is whether a colleague could pick it up in six months and reconstruct what you meant, which is the same test accountability sets.

And the last one matters most for this paper. **Writing a specification does not retain agency. It only creates the place where agency can be exercised.** If the specification itself is generated, skimmed and approved, you have not removed the handover. You have moved it earlier and made it harder to see, which is worse than where you started, because now there is a document that looks like a decision was taken.

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15. The counter-argument

A paper arguing for honest evidence has to state the case against itself, at full strength, and concede the parts that land. There are four objections I take seriously.

Agency is not intrinsically valuable. We gave up authorship of machine code, then memory management, then build pipelines, and nobody mourns any of it. Compilers removed an enormous amount of agency from programmers and the profession is unambiguously better for it. Assembly programmers at the time worried, correctly, that the next generation would not understand the machine. They were right, and it did not matter. So why is this anxiety different from that one?

It might not be, and I want to be honest that I cannot fully rule that out. The difference I would stake it on is that the compiler's abstraction was sound. It was deterministic, verifiable, and correct by construction, which is why you genuinely never needed to look underneath it. This abstraction is probabilistic and leaky, it fails plausibly rather than loudly, and you remain accountable for what is underneath it. An abstraction you are answerable for is one you still have to be able to read. The second difference is what sits in the layer. A compiler abstracted a mechanism with no judgement in it, whereas what gets absorbed now includes decisions about what to build and which trade-off to take, which is section 8's whole argument. And the concession is real: a great deal of what is being automated should be, and this paper is not a defence of craft. The question is not whether to hand work over. It is whether you can tell which handovers are mechanism and which are judgement, which is precisely what section 12 exists to answer.

Friction is not a virtue. Human checkpoints can be gatekeeping with better PR. Some of the agency being removed was being exercised badly: slowly, inconsistently, and with worse outcomes than the automated alternative. Anyone who has watched a change advisory board can see human judgement being applied in exactly the form this paper praises, and making things worse.

I concede most of this. The paper does not argue for more checkpoints, and section 12 says explicitly that every level is legitimate and that treating the most automated one as the destination is the failure. That cuts both ways. Treating the most supervised one as the safe default is the same failure wearing a high-visibility jacket. What I cannot give you is a way to tell your real judgement from your ceremony, because from the inside the two are indistinguishable, and a check that has become ritual still feels like diligence while it is happening.

The null result. Two years after ChatGPT launched, Danish administrative payroll data showed no detectable effect on earnings or hours, ruling out effects larger than 2% (Humlum and Vestergaard, 2026). If the economy-wide effect is that small, this may all be anticipation dressed as analysis.

It is a fair challenge and it deserves printing rather than managing. I could point out that Denmark has high union density and strong employment protection, and that earnings and hours are lagging and sticky, and both are true. But reaching for the methodology whenever the finding is inconvenient is the exact behaviour section 11 spends a section criticising, so I will not lean on it. The honest position is that the null constrains what I am allowed to claim. I am not claiming economic damage, and this paper would be wrong to. The claim is about where decisions get made and who can still make them, which payroll data cannot see in either direction. If in five years the structure of the work has changed as much as I think it has and nothing bad follows from it, I was wrong.

Recoverability. I have asserted throughout that agency loss is harder to reverse than productivity loss, and that is an argument rather than a finding. The argument runs like this: productivity is a capacity

you can rebuild by hiring, buying or spending, whereas judgement is accumulated through participation over time, and section 6 is about the route to that participation being removed. Rebuilding it needs the conditions recreated, not just the budget restored. But I have no evidence that this is true, and the falsifiable version is easy enough to state. If a team that has delegated fully for two years can be returned to confident unaided judgement in a quarter, the claim fails and most of Part I goes with it.

Which leaves the question worth asking of any paper like this: what would change my mind? Evidence that juniors reach competence as reliably under heavy delegation as they did without it. Evidence that teams who skip the specification arrive at the same decisions as teams who write one. And an organisation that has run a high-autonomy model for several years and can still reconstruct, from the record, why its systems do what they do. I have not seen any of the three, but I have not seen anyone look very hard either.

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16. Conclusion

This paper opened on a question most organisations cannot answer: when did you decide to let a machine make product decisions? The honest answer is that nobody did. That is not evasion and it is not poor record-keeping. The handover happened without a decision, and that is the only reason it could happen at all.

What happened instead is now easier to name. A licence rollout, a policy written before anyone had learned enough to know what it should say, a measure that rewarded acceptance over challenge, a ticket too thin to answer its own questions, an afternoon where it was faster to agree than to check. Every one of those is a reasonable act by a competent person. Together they are a handover, and nobody involved experienced it as one.

The four things do not go at the same speed. Authorship goes first and quietly, because nobody notices that the explanation has stopped existing until somebody asks for it. Judgement goes next, not through incompetence but because reviewing properly takes longer than accepting and the organisation is watching the clock. Control goes third, as options close through defaults, vendors and structure rather than decisions. **And accountability does not move at all.** That is the gap this paper has been describing, and it widens whether or not anyone is looking at it.

None of this is an argument for going slower. I think it is close to the opposite. The organisations that keep their agency will move faster, and not for sentimental reasons: they will know which handovers are mechanism and which are judgement, they will not spend a quarter reconstructing reasoning that was never written down, and they will still have people who can tell when the plausible answer is the wrong one. Being able to explain your own system is not a brake. It is what lets you change it with any confidence.

Which brings it back to the claim the middle of this paper rests on. **Agency does not have to disappear. But it does have to move.** It moves earlier, out of the review and into the intent, because

by the time something has been generated the decisions it contains have already been taken. That is what a specification is for, and it is why section 14 ends on a warning rather than a recommendation: writing one does not retain your agency, it only creates the place where your agency can be exercised. Something still has to be exercised there, by someone who has formed a view worth having. And views like that are built by people allowed to experiment against real work, get it wrong in a context that matters, and work out for themselves what good and bad look like in their own domain. Which is precisely the freedom most organisations removed first, in the name of safety, before anyone had learned anything.

If you want somewhere to start, keep it small enough to be real. Name the delegation level on one piece of work and say it out loud in the pull request. Take one recent change nobody can fully explain and time how long it takes to reconstruct why it is the way it is, because that number is your actual position. Then make assumptions a required output for a month and see what surfaces.

I am not writing this from the far side of the problem. I measured velocity and got autopilot, ran workshops that taught operation and called it capability, and worked out what I had done considerably later than I should have. Most of what is in here is a description of things I got wrong, arranged so that someone else might get them wrong less expensively. Nobody has finished solving this, and anyone who tells you otherwise is selling a harness.

But the decision is being taken in your organisation either way, this quarter, by default if not deliberately.

Nobody is going to ask you to sign this one. Which is exactly why it is still yours to make.

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